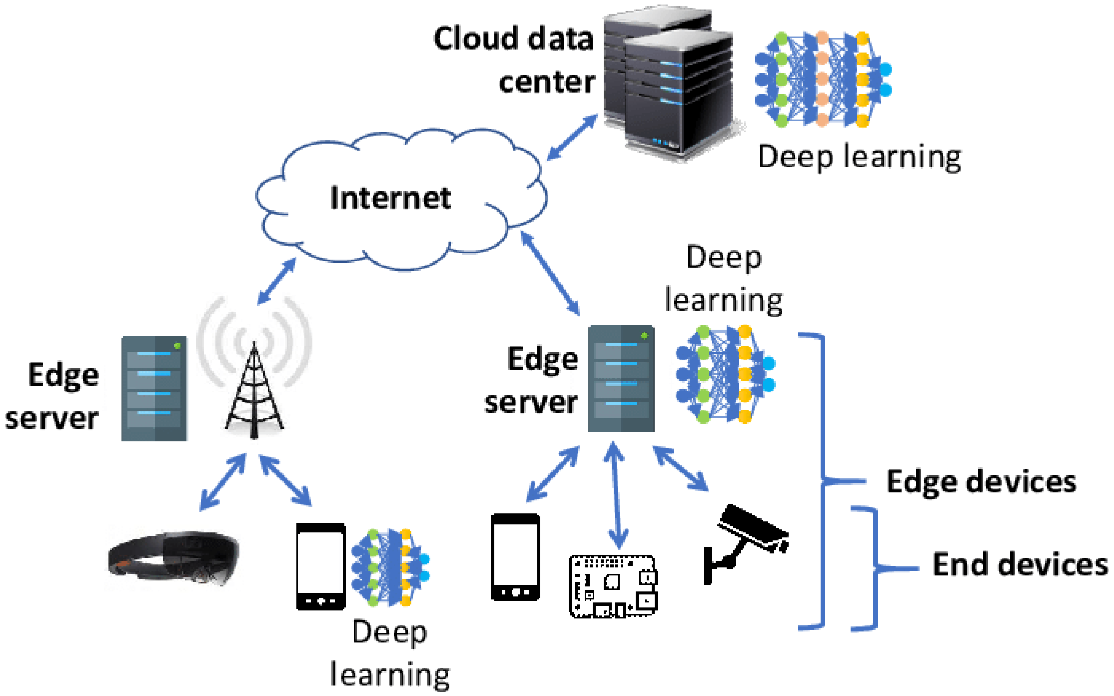
Device & Edge Security



Why This Layer is Vulnerable

- Physically accessible & remotely deployed
- Hard to patch & update
- First entry point for attackers

Device Security Essentials

- Secure boot & trusted firmware
- Unique device identity
- Authentication & authorization
- Firmware integrity protection

Edge Security Protections

- Access control & workload isolation
- Data encryption (in transit & at rest)
- Secure processing at the edge

Core Challenges in Edge Security

Decentralized Nature

Unlike centralized cloud systems, edge devices lack a single perimeter, making them harder to protect.

Resource Constraints

Many devices have limited processing power and memory, which hinders the ability to run robust, traditional security software.

Vulnerability Exposure

Consumer-grade and industrial devices often have weak, hardcoded security, limited update capabilities, and are frequent targets for hijacking into botnets.

Physical Insecurity

Edge devices are often deployed in unmonitored, public, or remote locations, making them susceptible to tampering.

Strategic Defenses & Frameworks

Zero-Trust Security

- Never trust, always verify
- Continuous authentication & device validation

Secure Edge Processing

- Process data locally to reduce exposure
- Use lightweight encryption & secure protocols

Human Firewall

- Train users to detect phishing & deepfakes
- Verify unusual requests & system alerts

Compliance & Governance

- Align with global security frameworks
- Follow lifecycle security best practices

Layered defense ensures resilience across the AIoT ecosystem

Key Technologies for Securing the Edge

Secure Edge Processin
AWS IoT Greengrass

- Local processing & secure messaging
- Device management data filtering

Secrets & Device Authentication
HashiCorp Vault

- Secure key & credential storage
- Device identity & authentication

Secure Communication
StrongSwan VPN

- Encrypted tunnels for data in transit
- Protects remote edge connectivity

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